What Do Fourier Neural Operators Compute? A Causal Coupling Probe and a Péclet Law for Spectral Mixing

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Abstract

Fourier Neural Operators (FNOs) are widely used surrogates for PDE solution operators, yet what they internally *compute* remains opaque. A common intuition—reinforced by the structured-matrix view of operator learning—is that an FNO behaves like a *circulant* operator, diagonal in the Fourier basis. We test this intuition causally. We introduce a black-box *Fourier coupling probe*: a finite-difference Jacobian in the Fourier basis whose diagonal concentration (*diagonality*) measures how close a learned operator is to circulant, complemented by mode ablation and activation patching. Calibrated against a provably circulant control and shown robust to its hyperparameters, the probe reveals that trained FNOs are *not* circulant: on viscous Burgers, diagonality falls monotonically from 0.40 to 0.08 as the flow becomes transport-dominated. We explain this with a short theoretical result: linearizing the dynamics splits the operator's Jacobian into a diagonal diffusion term and a strictly off-diagonal advection term (a convolution by the velocity spectrum), whose mass grows as $\mathrm{Pe}^{1/2}$ with the Péclet number. We then ask, with statistical care, how faithfully a *learned* FNO reproduces this structure. Three findings survive detrending, multiple seeds, and bootstrap confidence intervals. (i) The FNO's coupling matrix matches the exact operator's *in pattern* (not magnitude) at fixed Péclet number, correlation 0.77–0.99 (a within-regime comparison, immune to the shared monotonic trend; replicated on the reference `neuraloperator` library); in Frobenius norm the FNO's Jacobian is in fact magnitude-inaccurate (an accurate map with an inaccurate linearization). (ii) The FNO nonetheless *systematically under-responds to transport*: its diagonality scales as $\mathrm{Pe}^{-0.54}$ versus the exact operator's $\mathrm{Pe}^{-0.69}$ in 1D (slope-difference CI excludes zero), with the same significant effect in 2D Navier–Stokes and *resolution-invariant* from 64^2 to 256^2 and *capacity-invariant* over a 200× parameter range; it also appears on a *non-advection* reaction–diffusion PDE, making it a *general* FNO bias toward under-representing off-diagonal coupling (universal in sign and structure, though PDE-dependent in magnitude), not an advection quirk. (iii) A causal intervention—ablation with a double dissociation, and patching the exact operator's off-diagonal—shows off-diagonal coupling *mediates* transport and localizes the deficiency: the FNO's off-diagonal (advective) coupling is under-scaled to 0.3–0.5× physical, the operator-level mechanism of the under-response. The under-response survives at the Nyquist mode limit (a genuine representation bias, not a truncation artifact), and the diagnosis is *actionable*: a small mechanism-motivated spectral ($\dot{H}^1$) loss repairs the coupling and improves rollout along

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a fidelity–stability frontier, revealing the under-scaled coupling to be partly-functional spectral damping. We are deliberately conservative about what the probe does *not* show: the raw correlation of diagonality-versus-Péclet curves is near-tautological (shared monotonicity) and we do not report it as evidence; after detrending, the *scalar* diagonality carries no residual signal; and an apparent over-mixing gap is confounded by an architectural "leakage" (0.26 on a provably circulant target), so we do not claim it is physical. All experiments run at single-GPU scale.

1 Introduction

Neural operators learn maps between function spaces and have become standard surrogates for PDE solution operators, with the Fourier Neural Operator (FNO) [1] the most widely used instance. A mature theory describes *what they can represent*: universal approximation between Sobolev spaces and favorable size bounds for elliptic and Navier–Stokes problems [2], capacity-based generalization bounds [3], and a structured-matrix view in which each architecture corresponds to a matrix class inherited from the PDE's Green's function (FNO $\leftrightarrow$ circulant) [4]. Far less is known about *what a trained operator actually computes*—the interpretability question. Existing analyses are largely descriptive (e.g. FNO spectral bias [5]); causal, mechanistic tools from mainstream deep learning have scarcely been ported to operators.

We ask a concrete, falsifiable question: **does a trained FNO behave like its idealized circulant (diagonal-in-Fourier) structure, and if not, by how much and why?** Our contributions:

1. **A causal Fourier coupling probe** (§3): a black-box, architecture-agnostic finite-difference Jacobian in the Fourier basis, summarized by a *diagonality* functional, with mode-ablation and activation-patching cross-checks. We calibrate it against a provably circulant operator and show robustness to its hyperparameters.
2. **A monotone law** (§4): on viscous Burgers, FNO diagonality falls monotonically with transport; the operator approximates advection by learning *off-diagonal* coupling, not by recruiting more modes.
3. **A theory for the off-diagonal structure** (§5): the linearized generator splits into diagonal diffusion and strictly off-diagonal advection (convolution by the velocity spectrum), with off-diagonal mass $\propto \mathrm{Pe}^{1/2}$. This is a robust property of the *exact* operator.
4. **A statistically careful account of FNO faithfulness** (§7): with multiple seeds and bootstrap CIs, (a) the FNO's coupling *pattern* matches the exact operator at fixed Péclet (ρ=0.77–0.99; replicated on `neuraloperator`), but (b) the FNO *under-responds to transport*—diagonality slope -0.54 vs. -0.69 (CI excludes 0), significant in 1D and 2D, invariant across resolution (64^2–256^2) and capacity (200×). We explicitly retire the near-tautological raw correlation and an architecture-confounded over-mixing gap (§11).
5. **Causal mechanism** (§8): ablation with a double dissociation and off-diagonal patching show off-diagonal coupling *mediates* transport (not merely correlates), and localize the deficiency—the FNO's advective coupling is under-scaled to 0.3–0.5× physical—turning the probe from sensitivity analysis into causal evidence. A Nyquist-limit control rules out mode truncation as the cause.
6. **Generality** (§9): the under-response also arises on a non-advection reaction–diffusion PDE, so it is a general FNO bias toward under-representing off-diagonal spectral coupling, not an advection-specific effect.

7. **An actionable consequence** (§10): a mechanism-motivated spectral loss repairs the coupling and improves autoregressive rollout (a clean 40% long-horizon gain at 64^2), tracing a fidelity–stability frontier on which the under-scaled coupling behaves as partly-functional spectral damping.

2 Related work

Operator theory. FNO universal approximation and size bounds [2]; the proof projects the target onto retained Fourier modes, directly motivating mode-level analysis. Generalization bounds expose how the retained-mode count controls capacity [3]. **Interpretability.** FNO spectral bias is characterized empirically [5]; sparse-autoencoder operators parameterize concepts as functions [6], the only operator-native mechanistic work. We add *causal* probing tied to a physical invariant. **Transformer operators.** Attention is a learnable kernel-integral operator [7], contrasting FNO's fixed Fourier kernel—the basis for our cross-architecture comparison.

3 The Fourier coupling probe

Let $G : \mathcal{X} \to \mathcal{Y}$ be a (learned or exact) solution operator on the periodic torus, and $\hat{x}, \hat{y}$ the Fourier coefficients of input and output. We probe the input–output sensitivity in the Fourier basis by finite differences: perturbing input mode j by ϵ and reading output mode i,

$$J_{ij} \;=\; \frac{1}{\epsilon}\,\mathbb{E}_x\big|\,[\widehat{G(x+\epsilon e_j)}]_i - [\widehat{G(x)}]_i\,\big|, \qquad i,j = 0,\ldots,M{-}1. \tag{1}$$

A translation-invariant (circulant) operator is exactly diagonal in this basis. We therefore summarize J by its *diagonality*

$$\operatorname{diag}(J) = \frac{\sum_k |J_{kk}|}{\sum_{i,j} |J_{ij}|} \in [0,1], \qquad \operatorname{diag}(J) = 1 \iff \text{circulant}. \tag{2}$$

We also report two scale-free variants: row-normalized diagonality (each output mode's incoming sensitivity normalized to one) and the *coupling spectral entropy* (mean over rows of the normalized Shannon entropy; low for circulant, high for spread coupling). The probe is black-box: it requires only forward evaluations of G, so it applies identically to FNOs, Transformer operators, and the exact numerical solver. As mechanistic cross-checks we add (i) *mode ablation*—zeroing a retained spectral coefficient in each layer and measuring the rise in error—and (ii) *activation patching*—copying a single mode's post-spectral-conv activation from a source input into a target forward pass.

Calibration and robustness. Figure 1 illustrates the probe on a Burgers-trained FNO: the coupling matrix has clear off-diagonal ridges (diagonality 0.49), and ablation/patching localize computation to a few low modes. A provably circulant control—an FNO trained on pure linear diffusion—scores the highest diagonality ($\approx$0.85 here, 0.74 in the accuracy-matched control of §6; both well below the circulant ideal of 1, since non-spectral skip/MLP paths impose an architectural ceiling), versus 0.08–0.40 for Burgers (Fig. 2). Diagonality is flat across the finite-difference ϵ and its rank ordering is preserved across probe size, and the two scale-free metrics agree.

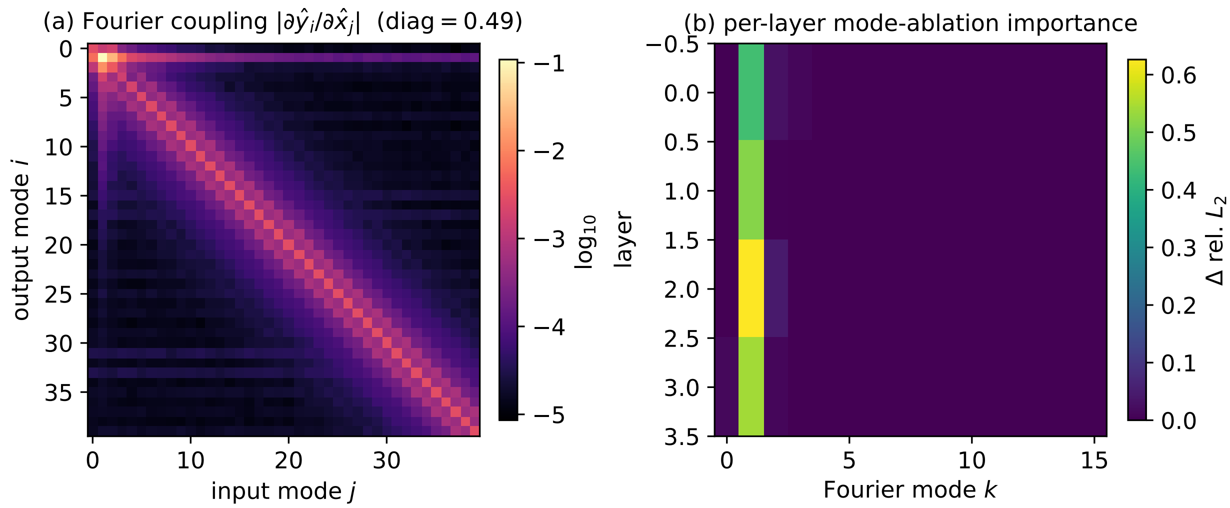

Figure 1: What the probe measures, on a Burgers-trained FNO (illustrative case, ν=0.05, probe band K=40; diagonality values are K- and ν-dependent and not compared across figures). (a) Fourier coupling matrix $|\partial\hat{y}_i/\partial\hat{x}_j|$ (log scale): off-diagonal ridges imply non-circulant structure. (b) Per-layer mode-ablation importance: computation concentrates in a few low modes.

4 A monotone law for spectral mixing

We train FNOs on viscous Burgers, $u_t = \nu u_{xx} - uu_x$, sweeping the viscosity ν from 5×10^{-2} (diffusion-dominated) to 5×10^{-3} (transport-dominated) with identical initial conditions. Figure 3(a) shows diagonality falling monotonically from 0.40 to 0.08. Strikingly, the FNO's *effective spectral rank* barely changes (Fig. 3(b)) while the data's spectral richness grows: the operator approximates transport not by recruiting more modes but by learning dense *off-diagonal* coupling within a small mode set.

5 Why: a Péclet law, and a falsifiable prediction

Linearize the dynamics around a background u^*; a perturbation δu evolves by $\partial_t \delta u = A\,\delta u$ with $A\,\delta u = \nu\,\partial_{xx}\delta u - \partial_x(u^*\delta u)$. In the Fourier basis,

$$\underbrace{(\nu\,\partial_{xx})_{kk} = -\nu k^2}_{\text{diagonal: diffusion}}, \qquad \underbrace{A^{\text{adv}}_{mk} = -\,i\,m\,\hat{u}^*_{m-k}}_{\text{off-diagonal: advection}}. \tag{3}$$

Because the mean is conserved and zero, $A^{\text{adv}}_{kk} = 0$: advection is *purely off-diagonal*, and its entries are set by the velocity spectrum. Constant advection ($u^* \equiv c$) leaves only $\hat{u}^*_0$, hence is still circulant; off-diagonal coupling requires a *spatially varying* velocity, which nonlinearity produces. To leading order in the horizon T, the off-diagonal mass of the Jacobian $J = \Phi_T$ is

$$\mathrm{OffDiag}(J)^2 \;\approx\; T^2 \sum_m m^2 \sum_{j\neq 0} |\hat{u}^*_j|^2 \;=\; T^2\Big(\textstyle\sum_m m^2\Big) E^*_{\text{ac}}. \tag{4}$$

Proposition 1 (Péclet scaling, informal)**.** *For Burgers with Péclet number* $\mathrm{Pe} = U/\nu$*, the solution spectrum spreads to* $k_\nu \sim \mathrm{Pe}$ *with* $|\hat{u}^*_k|^2 \sim k^{-2}$*; substituting into* (4)*, the off-diagonal mass grows*

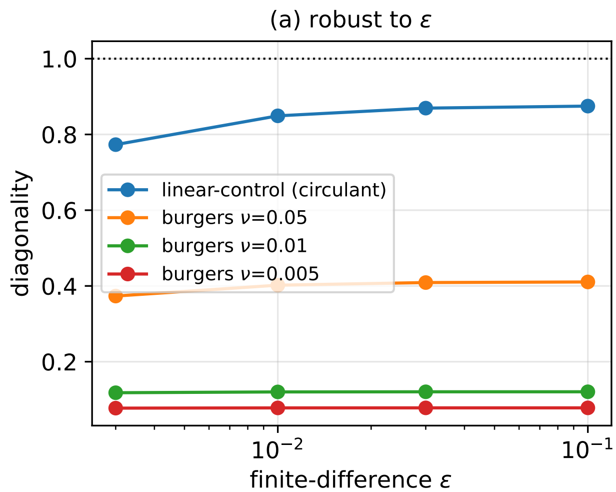

Figure 2: Calibration and robustness. The provably circulant linear-diffusion control scores highest; diagonality is (a) flat in ϵ and (b) order-preserving in probe size. The transport-dominated operators are unambiguously non-circulant.

monotonically with Pe*, so* diag(J) decreases monotonically with Pe*. Empirically the predicted mass scales as* $\mathrm{Pe}^{1/2}$ *(Fig. 4(b)).*

The off-diagonal coupling is therefore *forced by the physics*: it is a robust property of the exact operator, which we confirm by applying the identical probe to the exact spectral solver (Fig. 4). How faithfully a *learned* FNO reproduces it is the question we answer with statistical care in §7. We caution here that the raw correlation of the diagonality-versus-Pe curves is near-tautological—two monotone curves over a shared control parameter—so we do not treat it as evidence; the informative statistics are the fixed-Pe coupling-pattern match and the transport-scaling slope, reported next.

6 Disentangling architecture from accuracy

Two controls guard against over-interpreting the probe.

Architectural leakage. An FNO is not a pure spectral multiplier: its pointwise skip and MLP/activation paths add broadband coupling. Training an FNO on a *provably circulant* target (linear diffusion) to near-zero error (rel. $L_2 = 0.003$), the probe returns diagonality only 0.74, not 1.0—a 0.26 architectural "leakage" (Fig. 5). Any apparent FNO over-mixing on Burgers ($\lesssim 0.1$) is smaller than, and confounded by, this leakage; we therefore do *not* claim a physical over-mixing effect.

Transformer vs. FNO, accuracy-controlled. We probe a Transformer neural operator (multi-head softmax attention—a learnable, data-dependent kernel) across a range of accuracies. On these benchmarks the Transformer *cannot reach FNO accuracy* (best rel. $L_2 = 0.06$ vs. FNO's 0.015), so a matched-accuracy comparison is not available. What we can say honestly: the Transformer's diagonality stays near 0.09—below both the FNO (~ 0.14) and the exact operator (0.16)—across its *entire* accuracy range (Fig. 5), i.e. it does not approach the FNO as it improves. This is *suggestive*

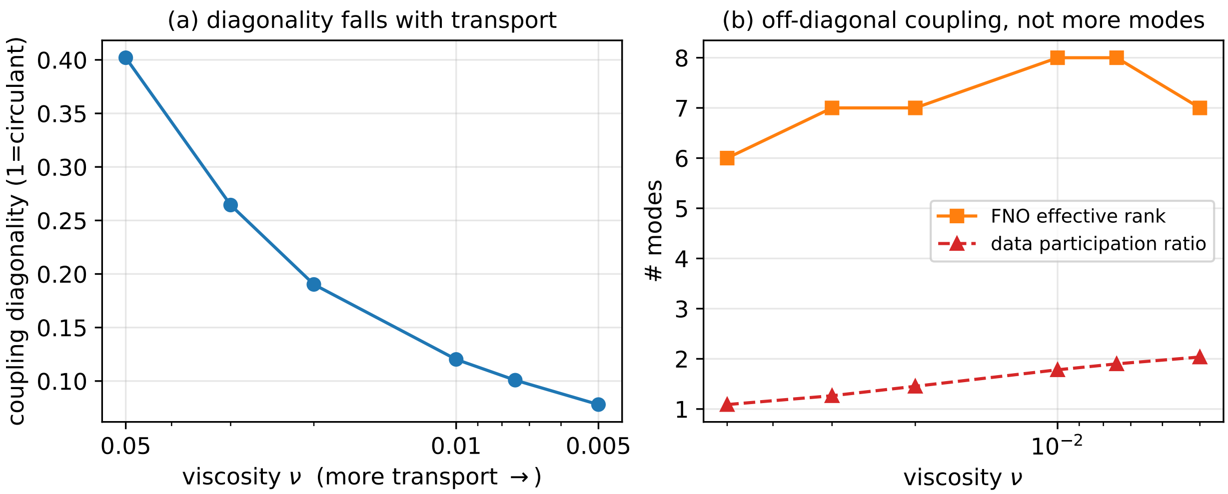

Figure 3: (a) Coupling diagonality falls monotonically as the flow becomes transport-dominated. (b) The effective spectral rank is nearly constant while the data participation ratio rises—off-diagonal coupling, not more modes.

of a genuine inductive-bias difference (a free kernel mixing more than a Fourier-parameterized one), but, lacking accuracy overlap, we report it as suggestive rather than established.

7 The main result: pattern match and transport under-response

We now quantify FNO faithfulness at scale: 2D Navier–Stokes (vorticity form, decaying turbulence) over a grid of **3 resolutions** $\{64^2, 128^2, 256^2\}$ × **7 Reynolds numbers** × **3 seeds** (63 trained models), with a float64 wavevector coupling probe. 2D advection $u \cdot \nabla\omega$ is again a convolution by the velocity spectrum, hence off-diagonal, so the theory predicts the same behavior; it holds, with two statistically robust findings (Fig. 7).

(1) Fixed-Péclet coupling-pattern match (pattern, not magnitude). Comparing the full coupling matrices $|J_{\mathrm{FNO}}|$ and $|J_{\mathrm{true}}|$ *at the same Reynolds number*—a within-regime test immune to the shared monotonic trend—the *pattern* correlation is $\rho = 0.77$ (low transport) to 0.99 (high transport), *invariant across resolution and seeds* (std ≤ 0.01), and it *replicates on the reference* `neuraloperator` *library* (Fig. 8). We stress this is a statement about the *shape* of the coupling, not its magnitude: in Frobenius norm the FNO's Jacobian differs substantially from the exact operator's (relative error 0.84–0.95), i.e. the FNO is an accurate *map* but its *linearization* is magnitude-inaccurate. §8 localizes that magnitude deficiency and shows it is causal.

(2) Transport under-response. The FNO nonetheless decreases its diagonality with transport *too slowly*: the exact operator scales as diag $\propto \mathrm{Pe}^{-0.42}$ while the FNO scales as $\mathrm{Pe}^{-0.35}$, a slope-difference of -0.068 with bootstrap CI $[-0.087, -0.051]$ (excludes zero), and an identical effect in 1D (-0.69 vs. -0.54, CI $[-0.20, -0.12]$). The consequence is a transport-dependent gap—FNO diagonality lies below the exact operator's at low Péclet (0.48 vs. 0.60 at $\nu{=}10^{-2}$) and converges at high Péclet—*not* a constant offset. The effect is **resolution-invariant** ($64^2/128^2/256^2$ agree to ~ 0.01; slope-difference $-0.068/-0.068/-0.072$) and **capacity-invariant**: over a $\sim 200\times$ parameter sweep (0.04–7.4M, 3 seeds) the diagonality varies by less than the across-seed std (e.g.

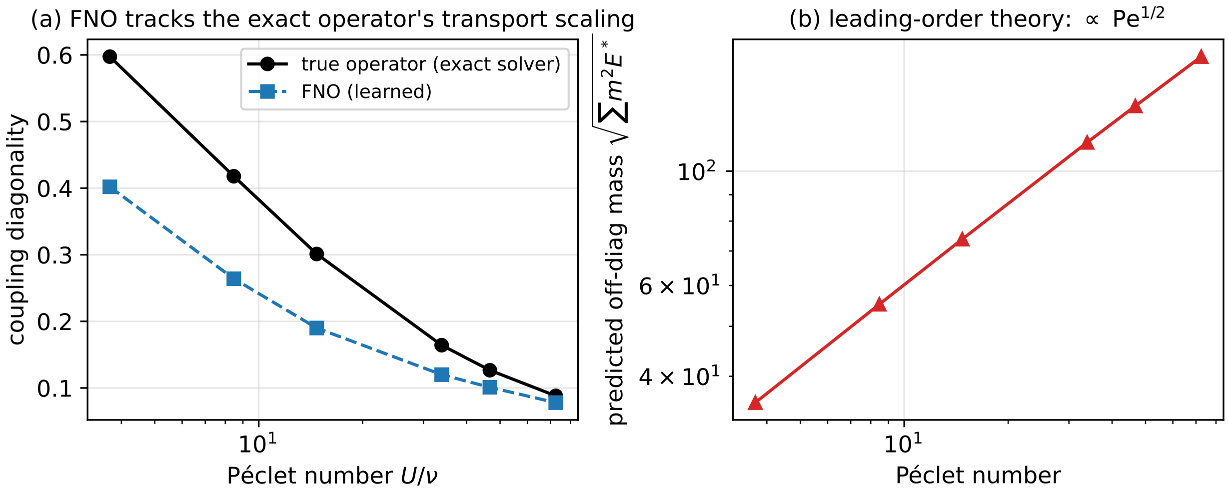

Figure 4: (a) The learned FNO's diagonality tracks the *exact* operator's across Péclet number (raw r shown only for reference; see §7 for the informative, detrended statistics). (b) The leading-order off-diagonal mass follows the predicted $\mathrm{Pe}^{1/2}$ law.

0.48–0.50 at ν=10^{-2}; pattern correlation 0.77 ± 0.005). After detrending against Pe, the residual of the *scalar* diagonality carries no significant signal (r=0.13, CI $[-0.18, 0.42]$); the under-response is a property of the trend (the slope), which is where we locate the finding. The slope-difference we report is at K=8; a probe-band spot-check (Appendix A) shows this is *conservative*—the effect is null at the lowest band and grows with K—so it is a mid/high-wavevector phenomenon, not an artifact of the low-K choice.

Not a mode-truncation artifact. An FNO retains a fixed number of Fourier modes and truncates the rest, so one might worry the under-response merely reflects an inability to represent coupling to truncated modes. It does not. Holding width and all else fixed, we sweep the retained-mode count up to the Nyquist limit—where *no* truncation occurs—and probe a fixed low band K=12 below the smallest retained count (Fig. 6). Both the off-diagonal under-scaling ($\|O_{\mathrm{FNO}}\|/\|O_{\mathrm{true}}\| \approx 0.49$) and the slope-difference (-0.059) are *flat across the sweep and identical at Nyquist*. The under-response is a genuine representation bias, not a consequence of mode truncation.

A precision caveat (methods lesson). The probe must use a cancellation-free Jacobian (autograd) or double precision. A float32 finite-difference suffers catastrophic cancellation—output coefficients are $O(10)$ while off-diagonal responses are $\sim 10^{-5}$, below the float32 floor—and spuriously reports near-uniform coupling (at K=8, ϵ=10^{-2}: float32 0.009 vs. float64 0.519 for the same model). We flag this for anyone building spectral-Jacobian probes at scale.

8 Causal mediation: off-diagonal coupling carries transport

The probe so far is a sensitivity analysis. We now make it causal by intervening on the operator's components. Working in the linear-response regime—where the generator split (diagonal = diffusion, off-diagonal = advection) is exact—we split both Jacobians into diagonal D and off-diagonal

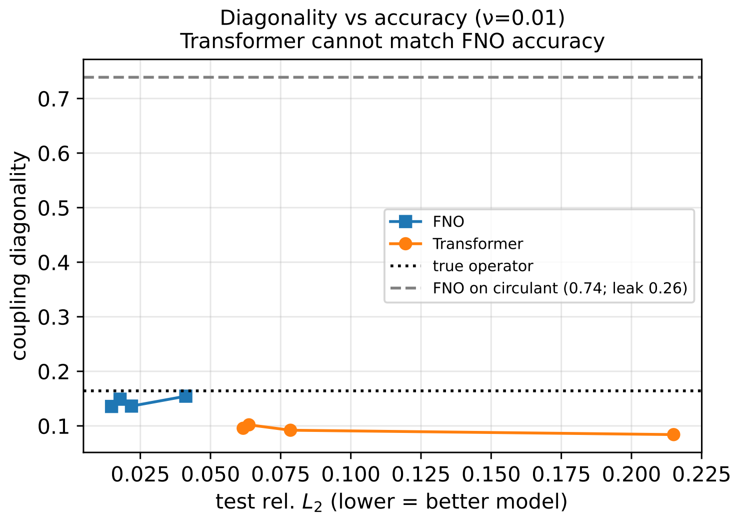

Figure 5: Diagonality vs. accuracy (ν=0.01). The Transformer cannot match FNO accuracy; its diagonality stays below the FNO and the true operator across its accuracy range (suggestive, not matched). Dashed line: an FNO on a circulant target reaches only 0.74 (0.26 architectural leakage), confounding any "over-mixing" interpretation.

O and measure the operator error $\|J - J_{\text{true}}\|_F / \|J_{\text{true}}\|_F$ under interventions, as a function of Péclet (Fig. 9; 7 Péclet $\times$ 3 seeds).

Double dissociation. Ablating the FNO's off-diagonal (D_{FNO}) raises error increasingly with Péclet (it is harmless at Pe≈4 but costs 0.14 at Pe≈98); ablating the diagonal (O_{FNO}) does the opposite, crippling the low-Péclet (diffusive) regime. Off-diagonal coupling is thus specifically implicated in transport, the diagonal in diffusion.

Patch recovery (mediation). Replacing the FNO's off-diagonal with the *exact* operator's ($D_{\text{FNO}} + O_{\text{true}}$) cuts operator error from 0.84 to 0.34 at high Péclet, with little effect at low Péclet: off-diagonal coupling is the channel that *mediates* transport accuracy, and the FNO's own off-diagonal is the dominant deficiency.

The deficiency, quantified. The FNO's off-diagonal magnitude is only $\|O_{\text{FNO}}\|/\|O_{\text{true}}\| =$ 0.29–0.47 of physical across Péclet (Fig. 9b). This under-scaled advective coupling is the operator-level mechanism of the transport under-response of §7.

Reconciling magnitude and ratio. Two facts might seem to conflict: the FNO's off-diagonal magnitude is *under*-scaled (this alone would raise diagonality), yet its diagonality *ratio* sits *below* the exact operator's (§7). Both hold because the FNO under-scales its *entire* linearization (Frobenius error 0.84–0.95), and under-scales the *diagonal even more* than the off-diagonal—so the ratio $\text{diag} = \|D\|/(\|D\| + \|O\|)$ drops even as $\|O\|$ itself falls. The two lenses are consistent: the FNO is uniformly magnitude-deficient, most severely on the diagonal, with the off-diagonal deficiency being the transport-mediating one.

9 Generality: a spectral bias beyond advection–diffusion

Is the under-response specific to advection, or a general property of how FNOs represent off-diagonal spectral coupling? We test a PDE with *no advection*: 1D Allen–Cahn reaction–diffusion,

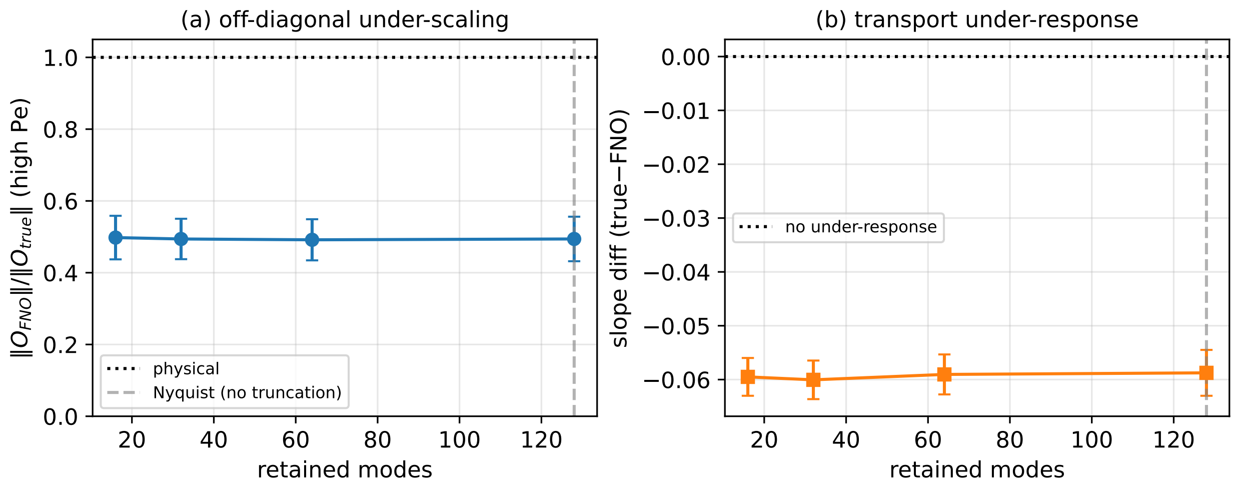

Figure 6: Mode-truncation control (1D, 3 seeds). Sweeping retained modes to the Nyquist limit (dashed line: zero truncation) leaves (a) the off-diagonal under-scaling and (b) the transport under-response slope-difference unchanged. The effect is not a truncation artifact.

$u_t = \varepsilon^2 u_{xx} + u - u^3$. Its off-diagonal coupling arises entirely from the cubic *reaction* (linearized: multiplication by $f'(u^*) = 1 - 3u^{*2}$, a convolution in Fourier), not transport. As the interface width ε shrinks, interfaces sharpen, the solution spectrum spreads, and reaction-induced off-diagonal coupling grows—the reaction–diffusion analog of increasing Péclet.

The under-response reappears (Fig. 10): the FNO's diagonality lies below the exact operator's, its slope is shallower, and the off-diagonal block is under-scaled ($\|O_{\text{FNO}}\|/\|O_{\text{true}}\| < 1$). Because the coupling here is reaction- rather than transport-induced, this identifies a **general FNO bias toward under-representing off-diagonal spectral coupling**, whatever its physical origin—and the generator-split framework extends accordingly (diagonal diffusion + off-diagonal reaction *or* advection).

Universal in sign and structure; magnitudes not comparable across PDEs. We are deliberate about the strength of this claim. What is *universal* across the two mechanisms is the *sign and structure*: the deficiency is always an *under*-scaling, always yields a shallower diagonality slope, and is always localized to the off-diagonal block. We do *not* compare slope-*magnitudes* across PDEs and do *not* claim reaction coupling is "more" suppressed than advection: the observed slopes (reaction ≈ -0.32; advection -0.15 in 1D, -0.068 in 2D) are measured over *different* control parameters (ε vs. Pe vs. Re), swept over *different* ranges (Appendix D) and mapped to diagonality by problem-dependent relations; a larger number can reflect a steeper control-to-diagonality map rather than a stronger bias. The shared, well-supported claim is strictly qualitative: an architecture-level under-representation of off-diagonal coupling, consistent in sign and structure. This mirrors our stance that the theory fixes the sign/monotonicity but not the empirical exponents (§11).

10 Closing the loop: the diagnosis is actionable

If the FNO under-scales advective (off-diagonal, high-frequency) coupling, a loss that up-weights high-frequency error should counteract it. We test this on an autoregressive setting: an FNO learns a one-step Burgers propagator $u(t) \mapsto u(t+\tau)$ at transport-dominated viscosity, trained with

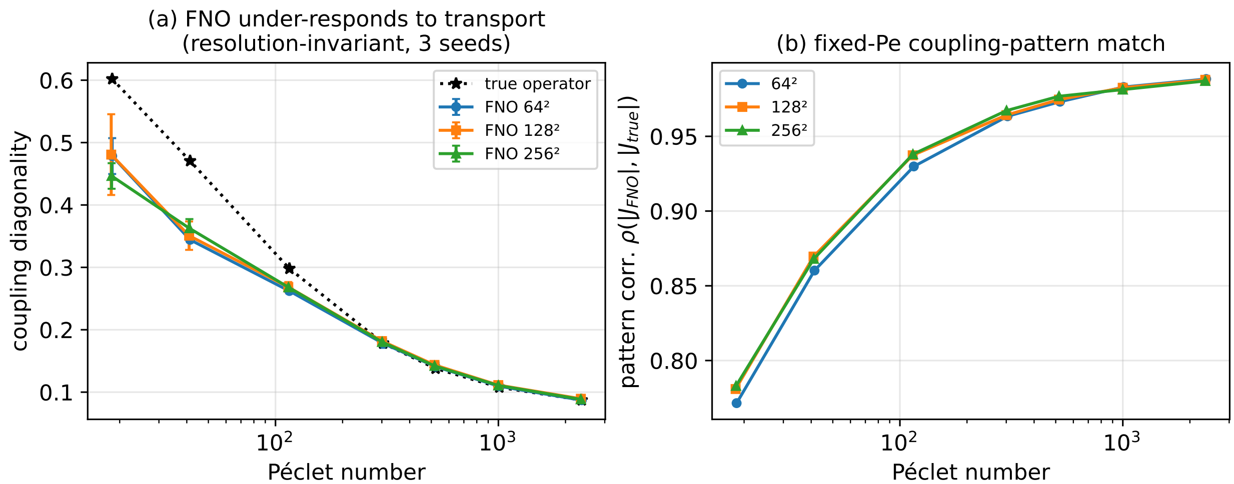

Figure 7: Scaled 2D study (63 models; error bars over 3 seeds). (a) The FNO under-responds to transport—below the exact operator at low Péclet, converging at high Péclet—*resolution-invariant* across 64^2–256^2. The near-overlap at high Péclet reflects the *conservative* $K{=}8$ probe band (Appendix A, Table 1); wider bands widen the gap, so this is a lower bound on the effect, not near-perfect agreement. (b) The fixed-Péclet coupling-pattern correlation (0.77–0.99), resolution- and seed-invariant.

MSE $+ \lambda \| \cdot \|^2_{\dot{H}^1}$ (a k^2-weighted spectral seminorm), then rolled out autoregressively for 20 steps (Fig. 12).

Sweeping λ traces a **fidelity–stability frontier**. A small weight ($\lambda \approx 10^{-5}$) is a sweet spot: it nudges the one-step operator's off-diagonal coupling toward physical ($\|O\|/\|O^*\|$ $0.92 \to 0.94$), improves one-step accuracy (4×, rel. L_2 $0.019 \to 0.005$) and short-horizon rollout ($0.034 \to 0.024$ at step 5), *without* sacrificing long-horizon stability (step 20: $0.134 \to 0.123$). Larger weights *over-repair*: they push the coupling closer to physical but long-horizon rollout error grows sharply ($0.13 \to 0.27$ by $\lambda{=}10^{-2}$). Thus the under-scaled advective coupling is partly *functional*—it acts as implicit spectral damping that stabilizes long rollouts, consistent with the known autoregressive instability of neural operators. The practical consequence: our diagnosis turns into a principled, mechanism-motivated knob—a small spectral loss improves fidelity while preserving stability, and the diagnosed deficiency should not be fully "corrected" away.

At practical 2D scale. The knob transfers to 2D Navier–Stokes (Fig. 11): a one-step vorticity propagator at 64^2 trained with the same $\dot{H}^1$ loss. Here the sweet spot ($\lambda{\approx}10^{-5}$) is a clean win—it drives the off-diagonal ratio to ≈ 1, improves one-step accuracy 5× ($0.025 \to 0.005$), and *reduces* long-horizon rollout error by 40% ($0.095 \to 0.057$ at step 15), with no over-repair penalty in the tested range. The mechanism-motivated fix is thus practically useful at scale, not just diagnostic.

11 Scope and limitations

Not generic Jacobian noise. That a network accurate in L^2 has an inaccurate Jacobian is, in isolation, unsurprising (adversarial sensitivity is an extreme case). Our claim is not that the FNO Jacobian is merely noisy: the error is *structured*—concentrated in the off-diagonal (advective) block, of a specific sign (under-scaling to 0.3–0.5×), monotone in Péclet, and *causally mediating* a

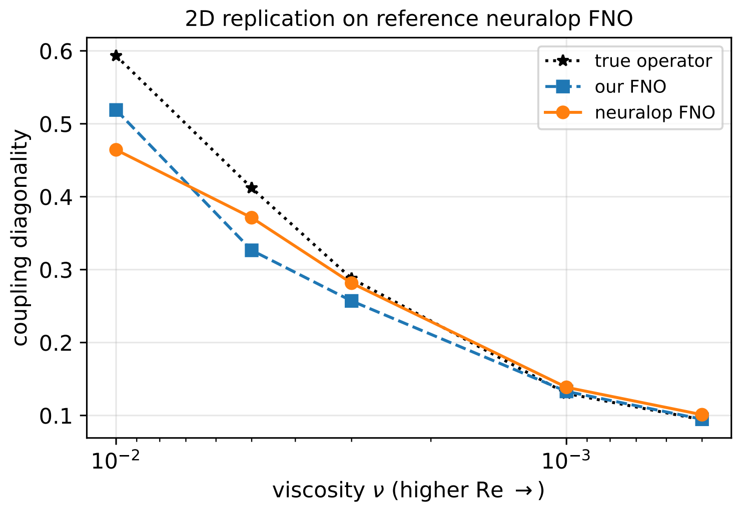

Figure 8: Replication on the reference `neuraloperator` 2D FNO: its coupling diagonality agrees with our FNO and with the exact operator at every Reynolds number (probed identically). The finding is not implementation-specific.

specific behaviour (transport) via ablation and patching (§8). Isotropic Jacobian noise would exhibit none of these. **Theory scope.** The generator split derives *why* the coupling is off-diagonal and that its mass grows monotonically with Péclet ($\propto \mathrm{Pe}^{1/2}$ to leading order, a short-horizon/frozen-coefficient expansion); it does *not* derive the empirical diagonality exponents ($-0.42/-0.35$ in 2D, $-0.69/-0.54$ in 1D), which we report as measured, not predicted. The theory motivates the probe and the diffusion/advection split; the exponents and the FNO deficit are empirical. **Precision & probe settings.** The probe requires a cancellation-free Jacobian (autograd or double precision; §7); the diagonality *value* depends on the probe band K and ϵ, so we fix them and use the linear-response limit. **PDE family.** We cover advection–diffusion (Burgers, Navier–Stokes) and reaction–diffusion (Allen–Cahn), spanning transport- and reaction-induced off-diagonal coupling; dispersive and hyperbolic operator classes remain open. **Baselines.** The Transformer operator does not reach FNO accuracy on these problems, so its lower diagonality is a suggestive inductive-bias diagnostic, not a matched-accuracy or SOTA comparison.

12 Conclusion

A simple, black-box causal probe shows that Fourier Neural Operators are not the circulant maps their idealized structure suggests: representing advection requires off-diagonal spectral coupling, a property we derive for the exact operator and that grows with the Péclet number. Treating FNO faithfulness as a statistical question, our findings survive seeds, bootstrap CIs, and replication on a reference library: the learned operator's coupling *pattern* matches the exact operator within each Péclet regime (ρ=0.77–0.99), yet the FNO *under-responds to transport*—its coupling decorrelates from advection too slowly (a significant slope difference, invariant across resolution 64^2–256^2, a 200× capacity range, and up to the Nyquist mode limit). The same signature—under-scaling, shallower slope, off-diagonal localization—recurs on a non-advection reaction–diffusion PDE, so the effect is an **architecture-level bias** toward under-representing off-diagonal spectral coupling, universal in sign and structure though PDE-dependent in magnitude. A causal intervention localizes it to

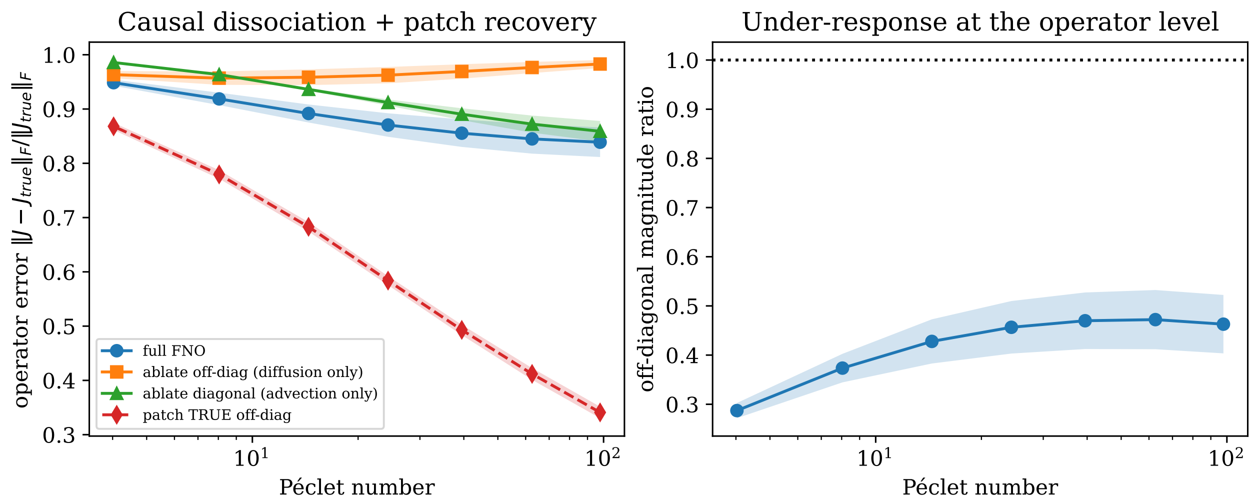

Figure 9: Causal mediation (linear-response regime). (a) Double dissociation: ablating off-diagonal coupling hurts transport (high Péclet), ablating the diagonal hurts diffusion (low Péclet); patching the *true* off-diagonal recovers high-Péclet accuracy. (b) The FNO's off-diagonal magnitude is systematically 0.3–0.5× the exact operator's—the mechanism of the transport under-response.

an under-scaled off-diagonal block, and a mechanism-motivated spectral loss turns the diagnosis into an actionable fidelity–stability knob (a clean 40% long-rollout gain at 64^2). We are explicit about what does *not* hold: the raw curve correlation is near-tautological, the detrended scalar residual is null, and an over-mixing gap is an architectural artifact. A methodological lesson—spectral-Jacobian probes require double precision or autograd to avoid catastrophic cancellation—emerged from scaling and is reported in full. The substantive interpretability question this opens is *when* a neural operator's internal computation tracks the PDE's spectral structure, and where it systematically departs; the probe is an instrument for studying it.

References

[1] Z. Li et al. *Fourier Neural Operator for Parametric Partial Differential Equations.* ICLR 2021.

[2] N. Kovachki, S. Lanthaler, S. Mishra. *On Universal Approximation and Error Bounds for Fourier Neural Operators.* JMLR 2021.

[3] T. Kim, M. Kang. *Bounding the Rademacher Complexity of Fourier Neural Operators.* Machine Learning, 2024.

[4] N. Boullé, A. Townsend. *A Mathematical Guide to Operator Learning.* Handbook of Numerical Analysis, vol. 25, 2024 (arXiv:2312.14688).

[5] S. Qin et al. *Toward a Better Understanding of Fourier Neural Operators from a Spectral Perspective.* arXiv:2404.07200, 2024.

[6] B. Tolooshams, A. Shen, A. Anandkumar. *Mechanistic Interpretability with Sparse Autoencoder Neural Operators.* arXiv:2509.03738, 2025.

[7] S. Cao. *Choose a Transformer: Fourier or Galerkin.* NeurIPS 2021.

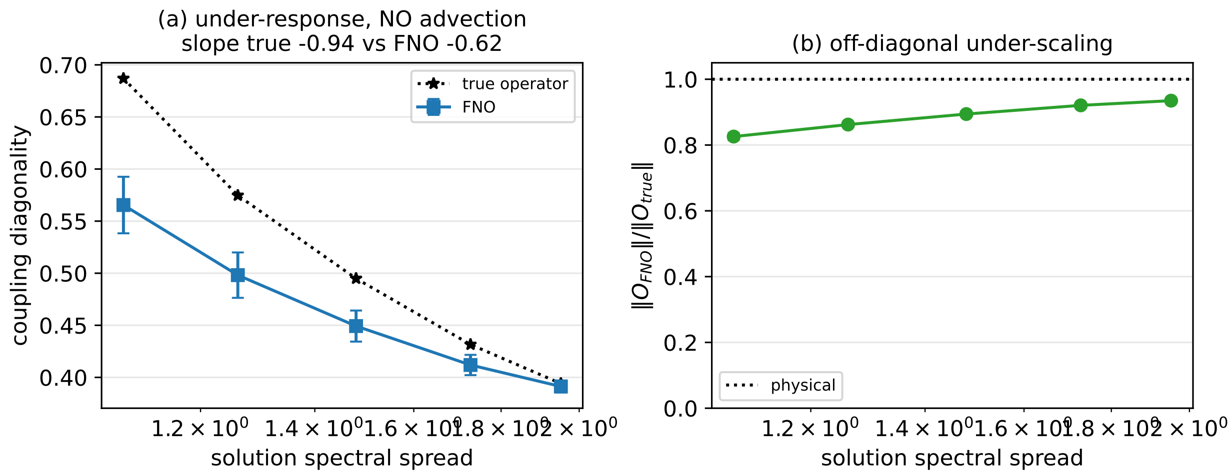

Figure 10: Generality on Allen–Cahn reaction–diffusion (no advection; 3 seeds). (a) The FNO under-responds—below the exact operator, shallower slope—as the reaction sharpens interfaces. (b) The off-diagonal (reaction) coupling is under-scaled. The under-response is not advection-specific.

A The coupling probe: definitions and precision

Finite-difference Fourier Jacobian. For an operator G on the periodic torus, we perturb input Fourier mode j by ϵ and read output mode i, averaging the magnitude response over a background set of inputs: $J_{ij} = \epsilon^{-1}\mathbb{E}_x\big|[\widehat{G(x+\epsilon e_j)}]_i - [\widehat{G(x)}]_i\big|$, for i, j in a probe band of K (1D) or $2K^2$ (2D, low wavevectors in rfft layout: rows $\{0..K-1\}\cup\{H-K..H-1\}$, cols $\{0..K-1\}$). Perturbing a single rfft coefficient and inverting yields a real field, so the probe is applied identically to FNOs, Transformer operators, and the exact solver (black box). Unless stated, background set size $N_b \in [8, 16]$.

Metrics. Diagonality $\mathrm{diag}(J) = \sum_k |J_{kk}| / \sum_{ij} |J_{ij}|$; row-normalized diagonality (rows scaled to sum 1, then mean diagonal); coupling spectral entropy (mean over rows of the normalized Shannon entropy of the row-normalized $|J|$). For the causal analysis we split $J = D + O$ (diagonal / off-diagonal) and report $\|O\|_F$ and operator error $\|J - J_{\text{true}}\|_F / \|J_{\text{true}}\|_F$.

Precision (critical). The finite difference must be evaluated in `float64` (or replaced by an exact autograd Jacobian): in `float32` the off-diagonal responses ($\sim 10^{-5}$) fall below the round-off floor of the $O(10)$ output coefficients, causing catastrophic cancellation and a spurious near-uniform J (§7). All reported Jacobians are `float64` finite-difference (models cast to double, including complex spectral weights) or `float64` autograd; we verified the 1D results are precision-insensitive and the 2D results require double precision. Finite-difference step $\epsilon \in [10^{-4}, 10^{-2}]$ (linear-response regime; diagonality is flat across this range in 1D). To be clear, `float32` is fatal *only* inside this finite difference—the subtraction of two $O(10)$ outputs differing by $\sim 10^{-5}$. It does *not* affect data generation or training, which involve no such cancellation; those run in `float32` throughout (the exact solver is re-run in `float64` only when it serves as the probe target).

Probe band K and cross-experiment comparison. The probe covers a low-wavevector band of size K (1D) or $2K^2$ (2D). Both the *absolute* diagonality and the *magnitude* of the slope-difference

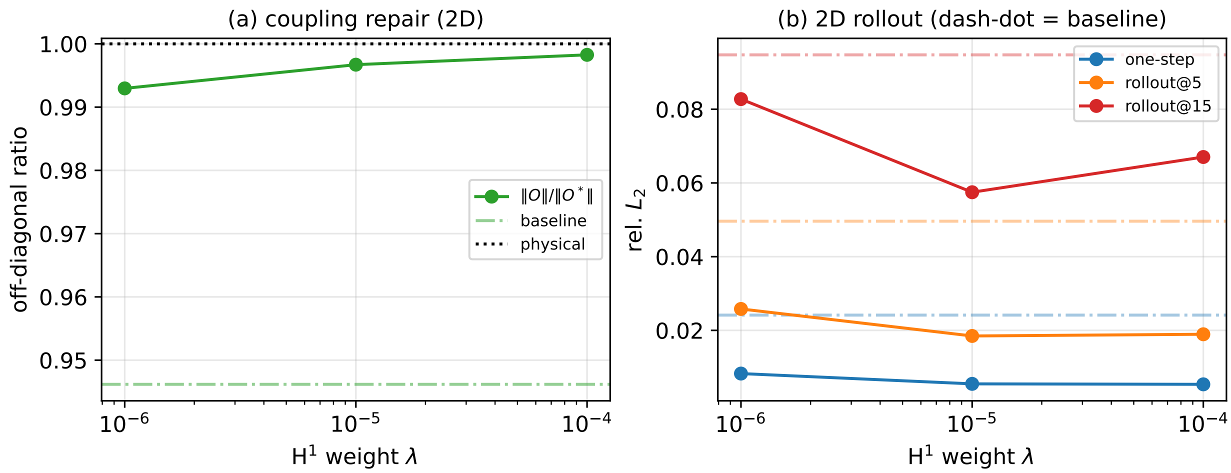

Figure 11: The knob at 2D scale (64^2 Navier–Stokes one-step propagator, 3 seeds; dash-dot = baseline). (a) The $\dot{H}^1$ loss repairs the off-diagonal coupling. (b) At $\lambda \approx 10^{-5}$ one-step and rollout error (through step 15) all improve over baseline. (Coupling ratios use the conservative K=8 band; cf. Table 1.)

depend on K, so **we compare only the *sign* of the under-response across experiments, never absolute diagonalities or slope magnitudes**; within-experiment magnitudes carry their own CIs. What is robust is the sign and structure (the FNO's slope is shallower and its off-diagonal under-scaled). A 2D spot-check at $K \in \{6, 8, 12, 16\}$ is revealing (Table 1): the under-response is *absent* at the lowest band (K=6: slope-difference -0.003 ± 0.015) and grows monotonically with K $(-0.05, -0.16, -0.25)$. It is thus a mid/high-wavevector phenomenon—the FNO reproduces the lowest modes' transport scaling and under-represents progressively at higher wavevectors—and our main-result choice K=8 (used because the `float64` full-Jacobian cost scales as K^2 at up to 256^2) is *conservative*: wider bands only strengthen the effect, so it is certainly not a low-band artifact.

K (band $2K^2$)	6 (72)	8 (128)	12 (288)	16 (512)
slope-diff (true−FNO)	−0.003	−0.049	−0.161	−0.251

Table 1: 2D K spot-check (64^2, 2 seeds): the transport under-response is null at the lowest band and grows with K; K=8 is conservative.

B PDE solvers

All solvers are pseudospectral with integrating-factor RK4 (IFRK4) time stepping and 2/3-rule dealiasing on the nonlinear term; the diffusive/linear part is integrated exactly via the integrating factor. Data are `float32`; the exact-operator probe re-runs the same solver in `float64`.

- **1D Burgers** $u_t + uu_x = \nu u_{xx}$, $[0, 1)$ periodic, 256 grid, $dt \in [10^{-4}, 2\times10^{-4}]$. Initial fields from a Gaussian random field with spectrum $(k^2 + \tau^2)^{-\alpha/2}$, τ=3, α=2.5, normalized to unit std.
- **2D Navier–Stokes** vorticity form, decaying turbulence, $\omega_t + u \cdot \nabla\omega = \nu\nabla^2\omega$, $u = \nabla^{\perp}\psi$, $\nabla^2\psi = -\omega$; $64^2/128^2/256^2$, $dt = 10^{-3}$. GRF vorticity (τ=5, α=2).

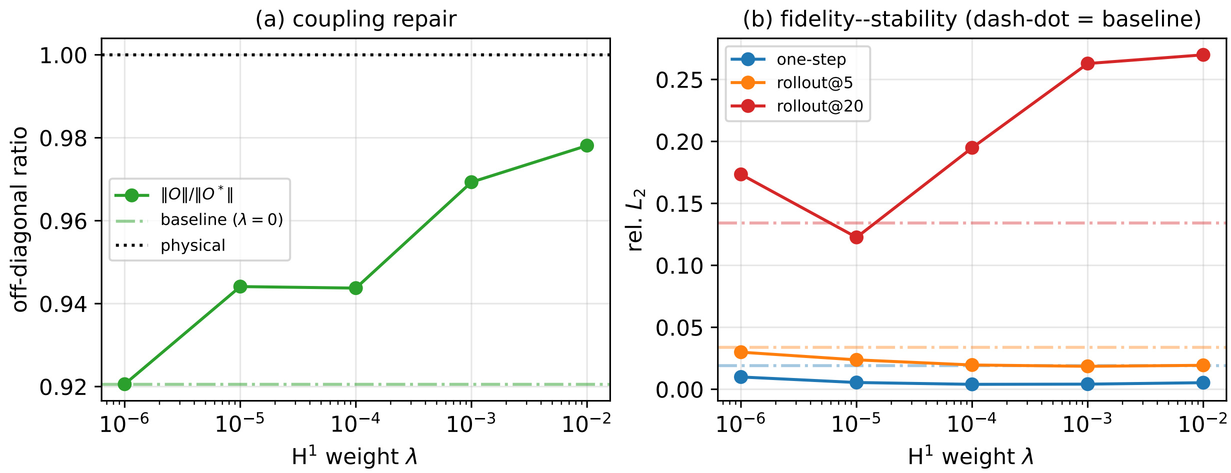

Figure 12: Closing the loop (1D one-step propagator, 3 seeds; dash-dot = untrained baseline λ=0). (a) An $\dot{H}^1$ spectral loss repairs the off-diagonal coupling as λ grows. (b) A sweet spot ($\lambda \approx 10^{-5}$) improves one-step and short-horizon accuracy without hurting step-20 rollout; over-repairing ($\lambda \gtrsim 10^{-3}$) trades long-horizon stability—the under-scaled coupling is partly functional damping.

- **Allen–Cahn** $u_t = \varepsilon^2 u_{xx} + u - u^3$, 256, $dt = 5{\times}10^{-4}$, $T = 1$, IC 0.5×GRF.
- **Linear diffusion** (circulant control): analytic multiplier $e^{-\nu k^2 T}$.

C Architectures and training

FNO (standard): lifting (input field + coordinates → width), 4 spectral-conv blocks with a pointwise skip and GELU, projection (width→128→1). 1D: width 64; modes as swept. 2D: width 32; modes 16/24/32 at $64^2/128^2/256^2$ (SpectralConv2d keeps two corner mode blocks). **Transformer operator**: pre-norm, 4-head softmax attention (learnable data-dependent kernel), model width 64, 4 layers, feed-forward 128, sinusoidal positional encoding. **Training**: Adam(W), lr 10^{-3} (Transformer $5{\times}10^{-4}$, grad-clip 1), weight decay 10^{-4}, cosine schedule, batch 20 (1D)/25 (2D, smaller at 256^2), 80–120 epochs, 600–1000 training samples. The closed-loop loss is MSE $+ \lambda\|\cdot\|^2_{\dot{H}^1}$ with the $\dot{H}^1$ seminorm $\sum_k k^2|\widehat{(\cdot)}_k|^2$.

D Per-experiment configuration

Experiment (fig)	PDE / dim	sweep	seeds	probe K, ϵ
Monotone law (2)	Burgers 1D	$\nu\, 0.05 \to 0.005$	1	$40, 10^{-2}$
Robustness (3)	Burgers 1D	ϵ, K grid + lin. ctrl	1	20/40/64
Theory (4)	Burgers 1D	6 ν	1	$40, 10^{-2}$
Main scaled (7,8)	NS 2D	$\{64, 128, 256\}^2 \times 7\,\mathrm{Re}$	3	$8, 10^{-3}$(f64)
Causal (10)	Burgers 1D	7 Pe	3	$40, 10^{-4}$(f64)
Capacity	NS 2D	(m, w) $0.04 \to 7.4$M	3	$8, 10^{-3}$(f64)
Mode-truncation (11)	Burgers 1D	modes $16 \to 128$	3	$12, 10^{-4}$(f64)
Accuracy-match (9)	Burgers 1D	FNO/Tr. families	1	$40, 10^{-2}$
Generality (13)	Allen–Cahn 1D	$\varepsilon\, 0.10 \to 0.025$	3	$12, 10^{-4}$(f64)
Closed-loop 1D (12)	Burgers 1D	$\lambda\, 0 \to 10^{-2}$	3	$12, 10^{-4}$(f64)
Closed-loop 2D (14)	NS 2D	$\lambda\, 0 \to 10^{-4}$	3	$8, 10^{-3}$(f64)

Reynolds sweep (2D): $\nu \in \{10^{-2}, 6, 3, 1.5, 1, 0.6, 0.3\} \times 10^{-3}$. Closed-loop propagators: 1D ν=5×10^{-3}, τ=0.05, spin-up t_0=0.2, 20 rollout steps; 2D ν=10^{-3}, τ=0.1, t_0=0.3, 15 steps.

E Statistics and compute

Confidence intervals are 95% bootstrap (5000 resamples) over the pooled (seed, control-parameter) runs; slope differences are fit in log–log (diagonality vs. Péclet / spectral spread). Reported $\pm$ are across-seed standard deviations. **Seed budget.** The headline claims—the transport under-response (main 2D result, causal mediation), its capacity- and resolution-invariance, generality on reaction–diffusion, and the closed-loop knob—use ≥ 3 seeds with bootstrap CIs (the 2D closed-loop uses 3; its 40% figure is a clean win *in the tested range*, not a variance-tight law). Calibration and illustrative experiments (the monotone-law figure, the ϵ/K robustness grid, the 1D theory overlay, and the accuracy-matched Transformer control) use a single seed; they support qualitative or sign-level statements, not the CI-backed headline numbers. All experiments run on a single NVIDIA A100 (40 GB); training in `float32`, probes in `float64`. Code: the probe (`fno1d.py`, `analysis.py`), solvers (`burgers_data.py`, `ns2d_gpu.py`, `rd_underresponse.py`), and one script per experiment; all figures regenerate from cached result JSONs via `make_figures.py`.